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Industrial Controls · Bay Area & Northern California

TRAINING PRIMER

Reading a 4-20 mA Loop: A Field Primer

Most of the analog signals moving around your plant ride on a 4-20 milliamp current loop. Level in a digester, pressure across a filter, the position of a modulating valve, the temperature of a bearing housing. If you can read one of these loops with a meter and turn the number into something real, you can settle a lot of arguments before anybody calls a vendor. This is a teaching primer, written the way I'd walk a maintenance tech through it standing at the panel. My goal is that by the end you understand *why* the signal works the way it does, not just which jack to plug the leads into.

Why this matters on the floor

Picture a level transmitter on a tank two hundred feet from the control room, wired back through a cable tray that also carries motor leads and VFD (variable frequency drive) output. The operator says the tank reads low. Is the tank actually low, is the sensor lying, or is the wire the problem? A 4-20 mA reading, taken right, tells you which world you're in. It's the single most useful measurement a controls person can teach a mechanical person to take, and it takes about five minutes once you understand it.

What an analog current loop actually is

An analog signal is one that varies smoothly to represent a real-world quantity, as opposed to a digital on/off. A **current loop** carries that analog signal as a flow of electrical current, measured in milliamps (mA, thousandths of an amp), around a single series circuit. The transmitter out at the equipment adjusts how much current it lets flow based on what it's measuring, and everything wired in series with it — the wire, the PLC (programmable logic controller) analog input card, any indicators — sees that exact same current.

That series arrangement is the whole trick. In a series circuit the current is identical at every point. So the number the transmitter *sends* is the number the input card *receives*, no matter what happens to the voltage along the way.

Why current and not voltage

Here's the part worth really understanding. If you sent an analog signal as a voltage — say 0 to 10 volts — you'd have a problem over any real distance. Wire has resistance. Push current through resistance and you lose some voltage along the way; that loss is called **voltage drop**. Two hundred

feet of small-gauge wire can drop enough that a 10-volt signal at the transmitter arrives as 9-point-something at the card, and now your reading is wrong and you don't know it.

Current doesn't care. In that series loop, whatever the transmitter sets, the same current flows everywhere, because the electrons have nowhere else to go. The wire resistance still causes a voltage drop, but the *current* — your signal — is untouched. As long as the power supply has enough voltage headroom to push the current through all that resistance, the reading holds.

Current loops also shrug off electrical noise better. A VFD or a big contactor throws electromagnetic interference that can induce stray voltage on a signal wire. A low-impedance current loop is much harder to nudge off its value than a high-impedance voltage signal. That's why in a plant full of motors and drives, 4-20 mA has been the workhorse for decades.

■ The live zero, and why it starts at 4

The signal runs 4 mA to 20 mA. Four milliamps represents 0% of whatever the instrument is scaled to measure, and 20 mA represents 100%. The range in between is linear, so 12 mA is 50%.

Why start at 4 and not 0? Because a **live zero** — a minimum signal that's still a real, flowing current — lets you tell the difference between "the process is genuinely at the bottom of its range" and "something is broken." Think it through. If 0% were 0 mA and a wire came loose, the loop would go to 0 mA and the system would read 0%. It would look like a perfectly valid empty tank. With a live zero, an empty tank reads 4 mA, and a broken wire reads 0 mA. Those are different numbers, and the control system can be told that anything at or near 0 mA means "fault," not "empty." That one design choice has prevented a lot of bad decisions.

There's a bonus: the 4 mA of standing current is enough to actually power the electronics in a simple transmitter, which leads us to how these things get their power.

■ 2-wire loop-powered vs 4-wire transmitters

A **2-wire transmitter**, also called loop-powered, is the common one. Two wires do everything: the same pair that carries the signal also carries the power. A DC power supply (typically 24 volts DC in industrial gear — check your equipment's spec) sits in the loop, and the transmitter works by *regulating how much current it draws* from that supply, from 4 mA up to 20 mA. Elegant: no separate power run, and the signal is the current draw itself.

A **4-wire transmitter** has its own power connection — two wires for power in, two for the signal out. You see this on instruments that need more juice than a loop can spare, like some analyzers, or gas detectors with heated sensors. The signal is still 4-20 mA and you read it the same way; it just isn't sipping its power from the loop.

Either way, three things share the loop: the power supply that pushes the current, the transmitter that sets the current, and a **burden resistor** (also called load resistance) somewhere — often the input of the PLC card, typically 250 ohms — across which the current develops a small voltage the card can

measure. The supply has to have enough voltage to cover the transmitter's minimum operating voltage plus the drop across all the wire and the burden. When people say a loop has "not enough compliance voltage," this is what starved it.

How to actually read the loop

You have three honest ways, and one common mistake to avoid.

Break the loop and put a DC milliamp meter in series. The signal is a series current, so to read it you make your meter part of the series path. Open the loop at a convenient terminal, set your multimeter to DC milliamps (mA, DC — not AC, not voltage), and connect it so the loop current now flows *through* the meter. The meter reads the loop current directly. Many field transmitters have a pair of test terminals or a HART jack exactly for this so you don't have to interrupt control. When you break the loop, the control system stops seeing its signal for that moment, so tell the operator first.

Use a true DC mA clamp. A clamp meter that specifically reads DC milliamps can measure the loop without breaking it — you clamp around one conductor. These are less common and pricier than ordinary clamps, and they need careful zeroing, but they let you read a live loop without opening it, which is a big deal on a running process.

Use a loop calibrator. This is a purpose-built tool that can read a loop in series, and can also *source* a known current (say, force 12 mA) to test how the PLC and the graphics respond, or power a transmitter on the bench. It's the right tool if you do this work regularly. One caution: sourcing a value into a loop that's still in service feeds the controller a number that isn't real, and the controller will act on it — it can start a pump, stroke a valve, trip an alarm. Before you source into a live loop, disconnect the transmitter or take that loop out of service, and put it back to normal when you're done. Never leave a control system running on a value you injected.

Now the mistake: **an ordinary AC clamp meter will not read a 4-20 mA loop.** The loop is direct current, DC, a steady flow in one direction. A standard clamp meter reads alternating current, AC, and will show you zero or garbage on a DC loop. Make sure you're on a DC milliamp function, whichever tool you use.

One more distinction that trips people up. If the transmitter has a HART display or a handheld shows a HART value, that number is what the transmitter *thinks* it's putting out — its internal digital value. The current your meter reads is what's *actually* flowing in the copper. Ninety-nine percent of the time they agree. When they *don't* agree — HART says 12 mA, your meter says 9 — you've just found something real: a scaling problem, a bad output stage, a ground path bleeding current. That disagreement is a gift. It tells you the fault is in the loop, not in the sensing.

NAMUR NE43: the fault bands

NAMUR is a European association of process-industry users that publishes widely-adopted instrumentation standards; its recommendation **NE43** defines how transmitters should signal a fault using the current itself. The idea: reserve the current values just outside the normal 4-20 range to mean "I'm broken or I'm out of range," so the control system can act on it.

Under NE43, the healthy live range is roughly 3.8 mA to 20.5 mA — a hair below 4 and a hair above 20, to allow for normal tolerance without crying wolf. Below about 3.6 mA, or above about 21 mA, the transmitter is declaring a fault: sensor failure, out-of-range, or electronics trouble. A well-set-up PLC input card knows these bands and can throw a diagnostic alarm instead of just clamping the reading. Exact edges vary slightly by manufacturer — check your equipment's spec — but the shape is always the same: a healthy band in the middle, fault bands biting in at each end, and 0 mA below all of it meaning the loop is dead open.

So when you meter a loop and see 3.5 mA, don't read it as "just under 0%." Read it as "this instrument is telling me it has a problem." Same with 21.5 mA at the top.

Turning milliamps into engineering units

The math is a straight-line conversion, and it's worth doing by hand a few times until it's second nature.

The signal spans 16 mA of range (20 minus 4) to cover 0% to 100%. So:

$$\text{Percent} = (\text{measured mA} - 4) \div 16 \times 100$$

Then apply that percent across the instrument's calibrated span in real units.

Worked example. Say you've got a pressure transmitter scaled 0 to 100 psi, and your meter reads 12 mA. Percent is $(12 - 4) \div 16 \times 100 = 50\%$. Fifty percent of a 0-to-100 psi span is 50 psi. Clean, because 12 is the middle.

Try one that isn't round. Same transmitter, meter reads 8 mA. Percent is $(8 - 4) \div 16 \times 100 = 25\%$. Twenty-five percent of 100 psi is 25 psi. And 16 mA gives $(16 - 4) \div 16 = 75\%$, or 75 psi.

If the range doesn't start at zero — say a temperature transmitter scaled 50 to 250 degrees F, a 200-degree span — you take the percent, apply it to the span, then add the low end. At 12 mA (50%): half of 200 is 100, plus the 50-degree low end equals 150 degrees F. The mistake to avoid is forgetting to add the offset when the range doesn't start at zero.

Common problems, at a teaching level

Open loop. A broken wire, a loose terminal, a corroded ferrule. The current stops. You'll read 0 mA, and NE43-aware systems flag it as a fault. This is the live zero earning its keep.

Wrong scaling. The loop is fine and reading, say, a true 12 mA, but the PLC is configured as if the transmitter were scaled differently than it actually is — 0-150 psi in the software when the transmitter is really 0-100. The current is honest; the interpretation is wrong. This is exactly the case where your metered mA disagrees with what the screen shows, and the mA is the truth.

Ground loop. When the loop gets connected to earth ground at more than one point, stray current can find a second path, and your reading drifts or reads high or wobbles. A loop should be grounded at one point only, usually at the supply or the card. Two grounds, two paths, trouble.

Moisture. Water in a junction box, a transmitter housing, or a connector creates leakage paths that bleed a little current where it shouldn't go. Readings drift, especially with weather or wash-down, and often come and go. On our kind of plants — wet, dusty, corrosive — this is one of the most common gremlins. Dry it out, reseal it, and look at whether the enclosure rating and the conduit drainage are right for where it lives.

| Safety, woven in

You're taking a live reading, which means the loop is energized. On a plain 24 VDC instrument loop the shock hazard is low, but treat every circuit as live until proven otherwise, keep to the one-hand-in-your-pocket habit around energized parts so you never bridge a circuit across your chest, and be aware of what *e/se* is in that enclosure. Panels often mix 24 VDC signal wiring with 120 or 480 VAC power in the same box. That AC is the real hazard, and opening a live panel puts you in arc-flash territory — wear the personal protective equipment (PPE) your facility's arc-flash study calls for, and follow your site's energized-work rules. And watch where the loop terminates: a 4-20 mA speed reference or speed feedback often lands on the terminals inside a variable frequency drive (VFD), and a VFD holds a lethal charge on its internal DC bus for several minutes after you kill power. Wait out the drive's rated bus-discharge time and verify the bus is down with a meter before you reach into that enclosure. If the job is to actually *work* on the loop, not just read it, follow lockout/tagout (LOTO) and get to a zero-energy state first.

One rule that is not negotiable. If the loop is part of an **intrinsically safe (IS)** circuit — a circuit deliberately limited in energy so it can't ignite a flammable atmosphere, common anywhere you have biogas, digesters, or Class I hazardous areas under NEC 500/505 — you do **not** break it or meter it with ordinary tools. Only approved, IS-rated instruments may enter that circuit, because a normal meter can introduce enough energy to defeat the whole protection scheme. Inserting a standard multimeter into an IS loop in a methane atmosphere is how you get an ignition source where the entire point was to have none. If you're not certain a loop is IS, find out before you open it, and remember biogas spaces bring their own hazards: methane is flammable between roughly 5% and 15% in air (its LEL and UEL), hydrogen sulfide (H₂S) is dangerous around 100 ppm (its IDLH level) and deadens your sense of smell so you can't trust your nose, and many of these are permit-required confined spaces that need a 4-gas monitor and a permit before entry.

Try this on your own plant (safely)

Pick one healthy, non-IS loop you already trust — a pressure or level transmitter feeding a good reading on the HMI (human-machine interface). With the operator's OK, find its test terminals, set a multimeter to DC mA, and read the loop current. Now do the math: convert your mA to percent, apply it across the transmitter's scaled range, and compare to what the screen shows. When your hand math matches the graphic, you've proven you understand the whole chain from copper to display. Do that on a known-good loop in daylight with no pressure on, and the day you need it on a suspect loop, it'll be muscle memory.

This pairs with the companion primer, "Is it the sensor or the process?" — once you can read the loop, that one helps you decide what the number is telling you.

Glossary

- **Analog signal** — a signal that varies smoothly to represent a real quantity, as opposed to digital on/off.
- **Current loop** — a series circuit that carries an analog signal as a flow of current; the same current flows at every point.
- **Milliamp (mA)** — one thousandth of an amp; the unit of the loop signal.
- **Voltage drop** — voltage lost across the resistance of the wire when current flows through it.
- **Live zero** — using 4 mA (not 0) for 0%, so a dead loop reading 0 mA is distinguishable from a genuine low reading.
- **2-wire (loop-powered) transmitter** — one that draws its power from the same two wires that carry its signal.
- **4-wire transmitter** — one with separate power wiring, apart from its signal pair.
- **Burden / load resistance** — the resistor (often the PLC input, typically 250 ohms) across which loop current makes a measurable voltage.
- **Compliance voltage** — the supply voltage headroom needed to push the loop current through all the resistance in the loop.
- **PLC (programmable logic controller)** — the industrial computer that reads the loop and runs control logic.
- **VFD (variable frequency drive)** — a motor drive; a common source of electrical noise near signal wiring.
- **NAMUR** — a process-industry user association that publishes instrumentation standards.
- **NE43** — the NAMUR recommendation defining fault-signaling current bands (healthy ~3.8-20.5 mA; fault at/below ~3.6 mA or at/above ~21 mA).
- **HART** — a digital protocol layered on the 4-20 mA loop; its reported value is the transmitter's internal number, not necessarily the measured loop current.

- **Loop calibrator** — a tool that reads a loop in series and can also source a known current for testing.
- **Ground loop** — an unintended second earth-ground path that corrupts the reading.
- **Intrinsically safe (IS)** — a circuit limited in energy so it can't ignite a flammable atmosphere; only IS-rated tools may enter it.
- **LOTO (lockout/tagout)** — the procedure to reach a zero-energy state before hands-on work.
- **PPE (personal protective equipment)** — protective gear, including arc-flash-rated equipment for energized panel work.
- **HMI (human-machine interface)** — the operator screen where the scaled reading is displayed.
- **LEL / UEL** — lower and upper explosive limits, the flammable concentration range of a gas in air.
- **IDLH** — immediately dangerous to life or health; an exposure threshold, ~100 ppm for H₂S.